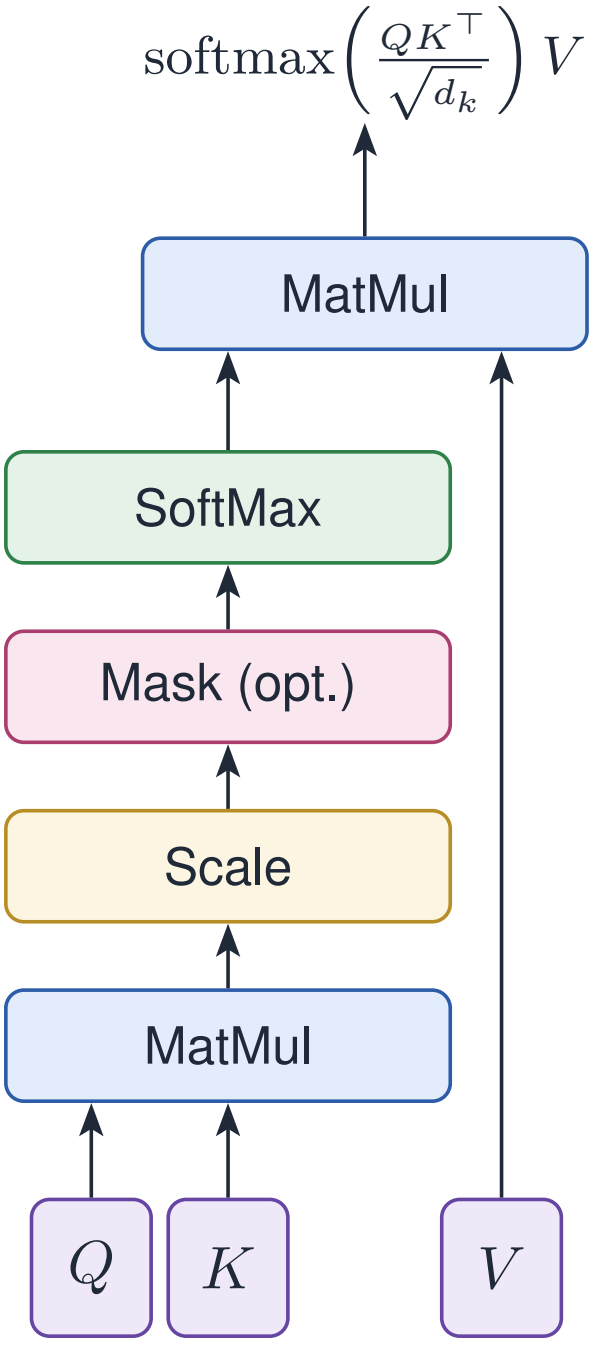
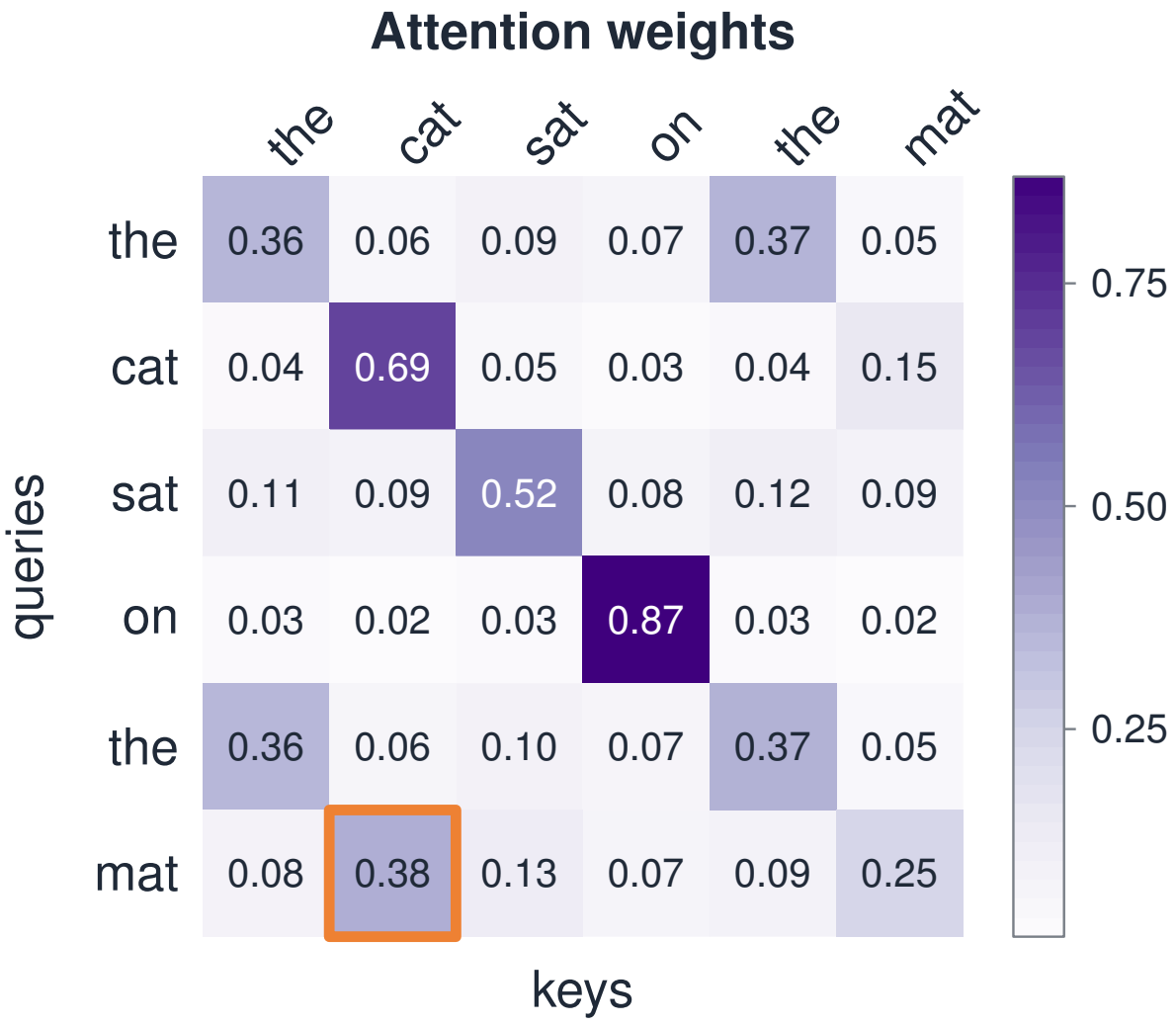

$\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$
MatMul
SoftMax
Mask (opt.)
Scale
MatMul
Q
K
V


Attention weights
the
cat
sat
on
the
mat
queries
the 0.36 0.06 0.09 0.07 0.37 0.05
cat 0.04 0.69 0.05 0.03 0.04 0.15
sat 0.11 0.09 0.52 0.08 0.12 0.09
on 0.03 0.02 0.03 0.87 0.03 0.02
the 0.36 0.06 0.10 0.07 0.37 0.05
mat 0.08 0.38 0.13 0.07 0.09 0.25
keys
0.75
0.50
0.25